# Bandgap Reference (BGR)


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# Design Specification

**Table 1:** Design specification of the bandgap reference

| Parameter | Specification |
|---|---|
| VDD, VSS | 1.8V, 0V |
| Vref | 1.0V |
| Temp. Coefficient(Tc) | <30ppm/°C |
| Power | <200uW |
| Input Pin | 2 |
| Output Pin | 1 |

# TB_BGR_Top

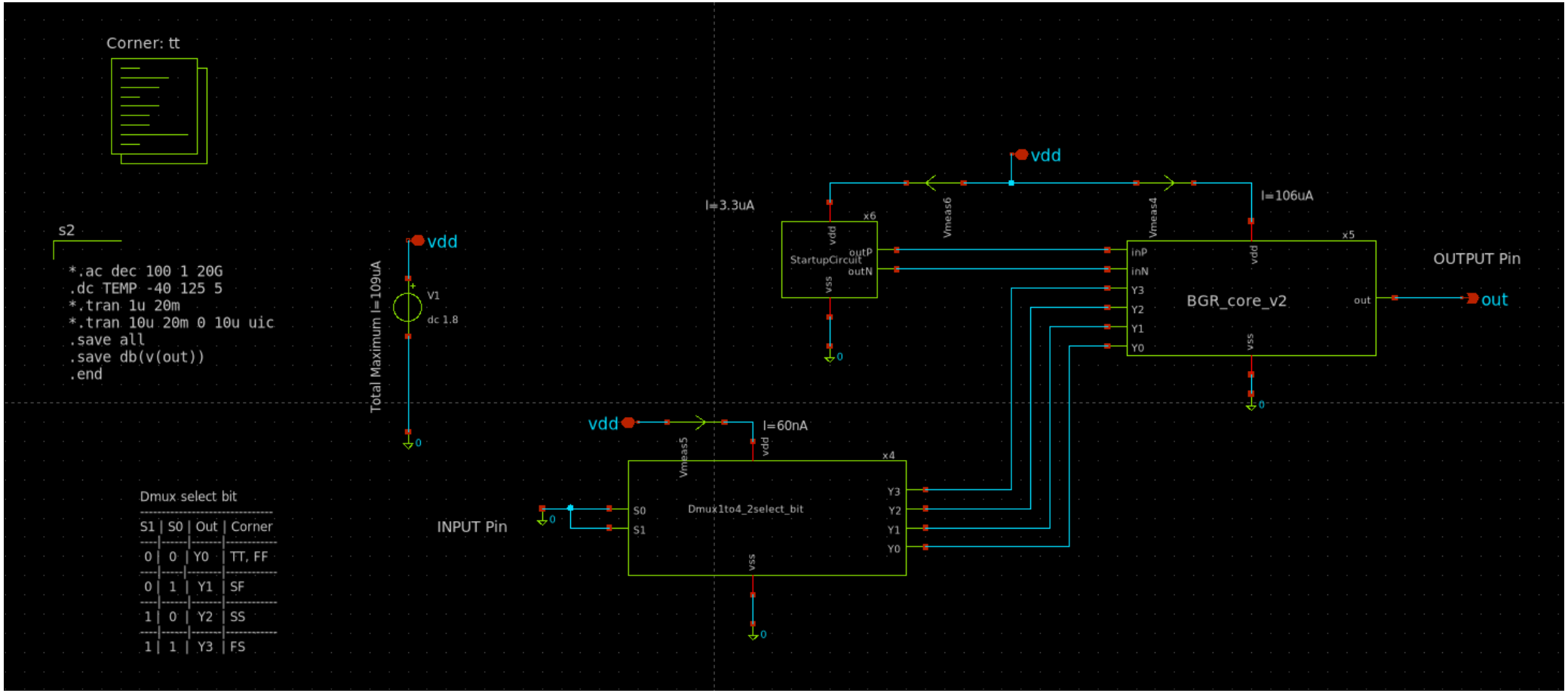



**Figure 1:** Block-level top testbench (TB_BGR_Top)

# TB_BGR_Top

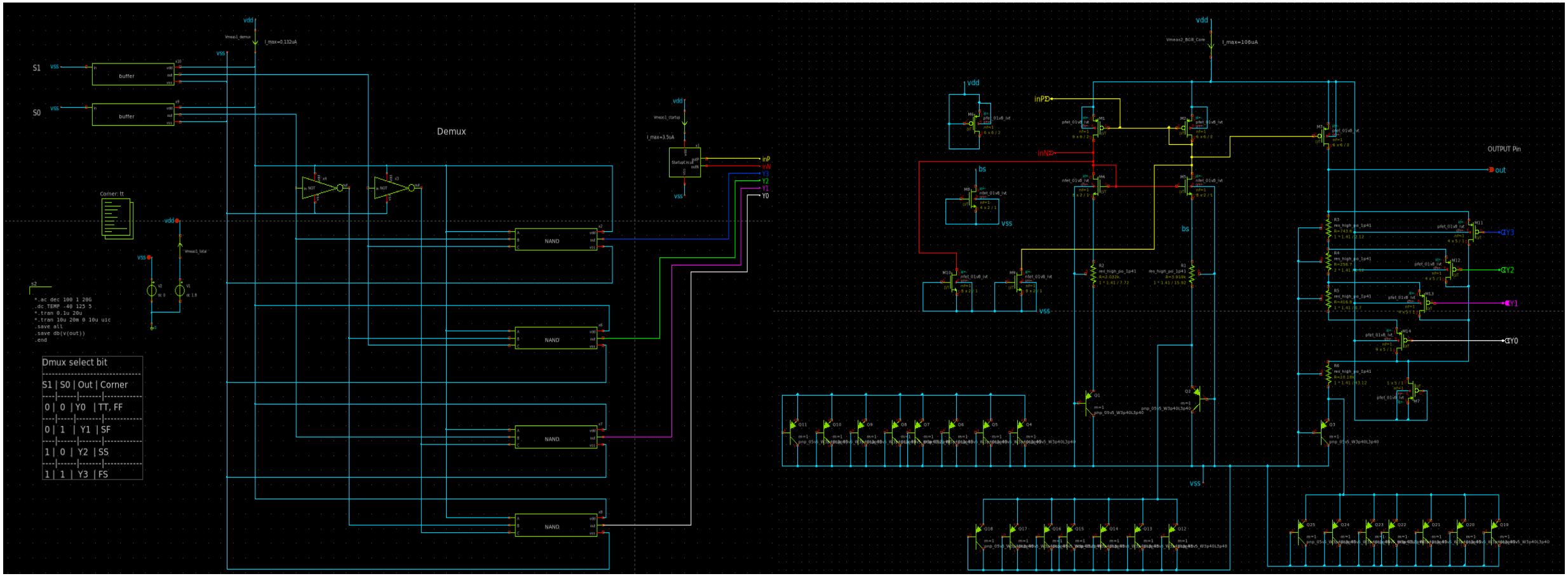



**Figure 2:** Device-level top testbench (TB_BGR_Top) with the BGR core built from primitive devices

Note: Block level Testbench shows different Temperature coefficient (TC) result so another TB is created by making BGR core in device level.

# Startup Circuit

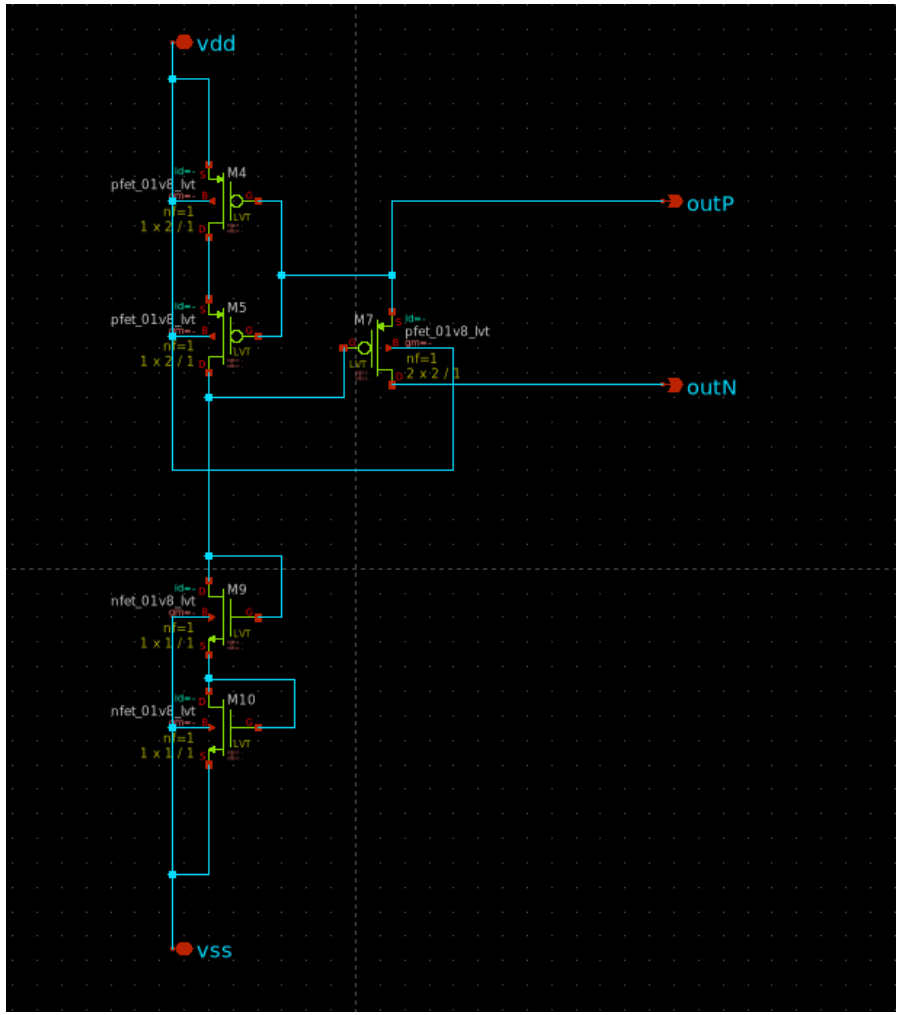
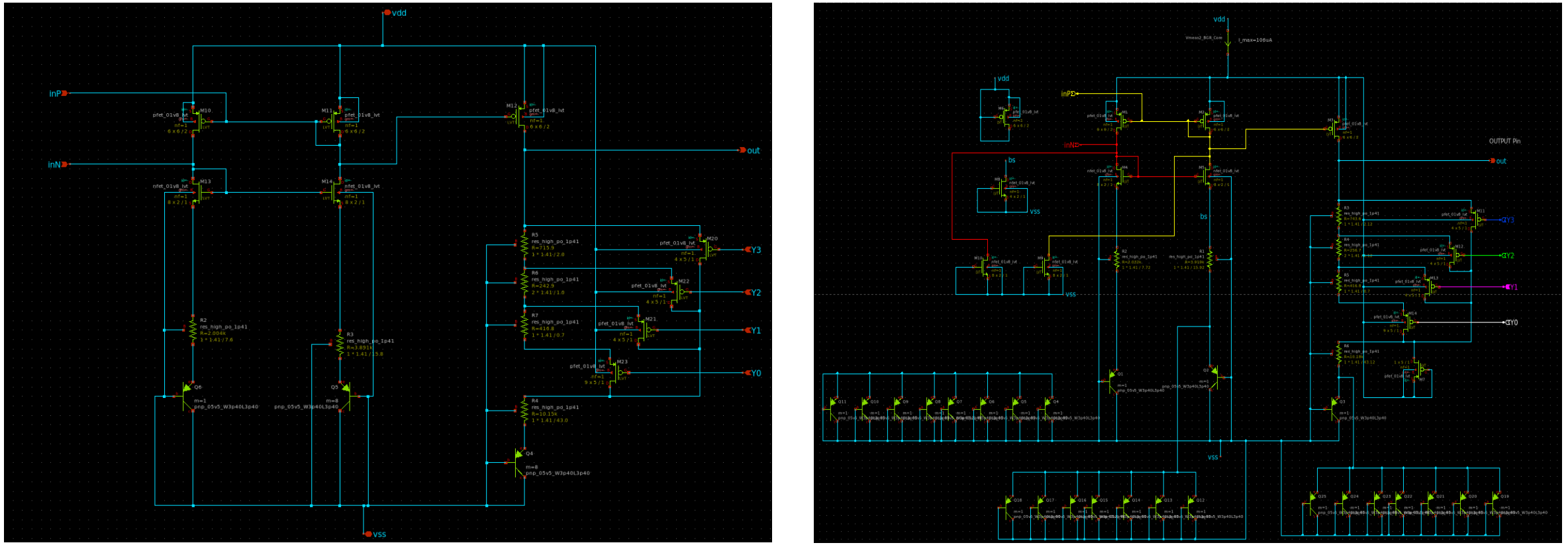



**Figure 3:** Startup circuit schematic

# BGR Core and Vref Branch




*(a)* *(b)*

**Figure 4:** BGR core and Vref branch: (a) schematic, (b) BJT multipliers drawn in parallel for layout

Note: For layout, all BJT's multipliers are shown in parallel.

# Dmux Schematic

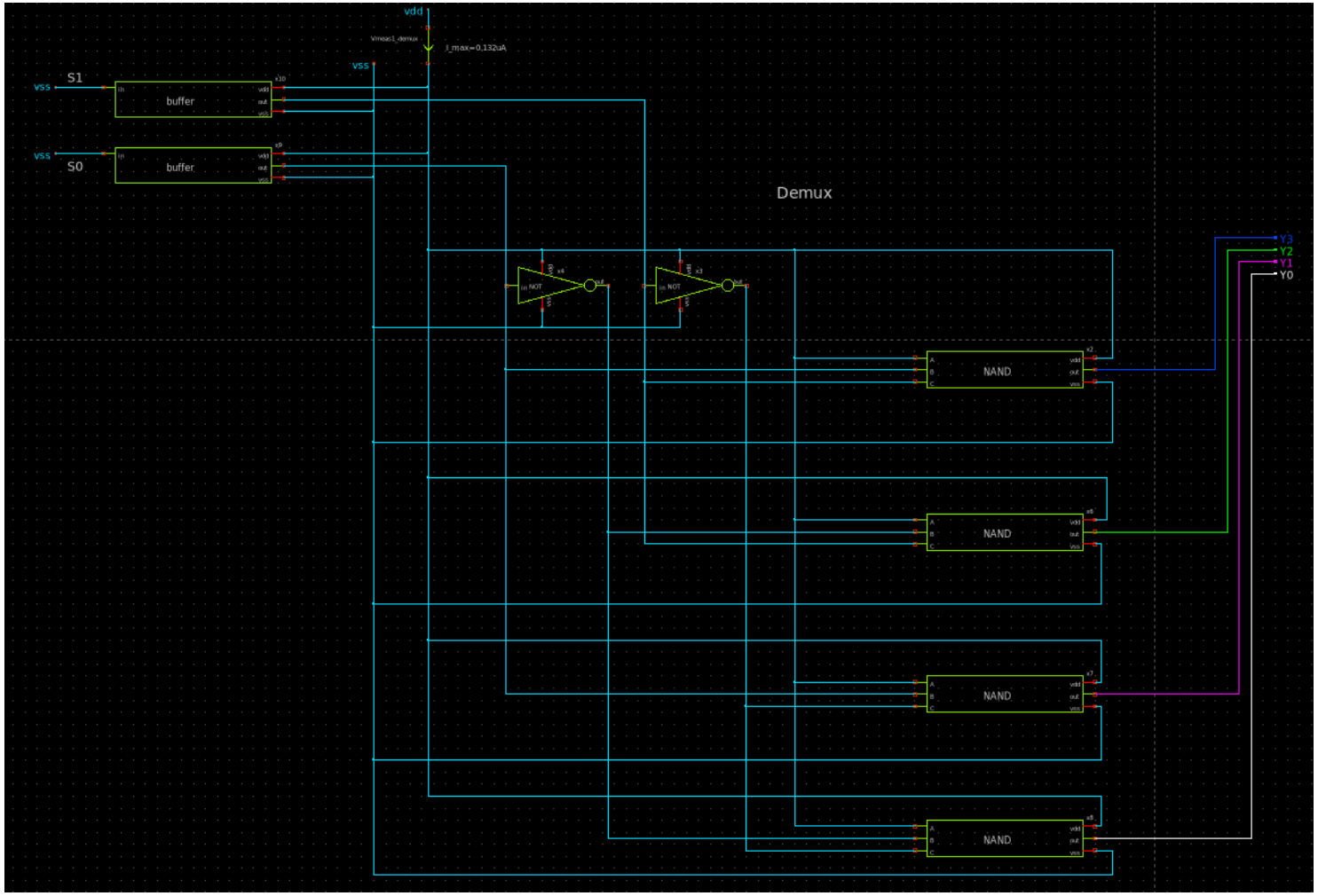



**Figure 5:** Dmux schematic

# Dmux select bit

**Table 2:** Dmux select-bit settings and the corresponding process corners

| S1 | S0 | Y3 | Y2 | Y1 | Y0 | Corner |
|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 | 0 | TT, FF |
| 0 | 1 | 1 | 1 | 0 | 1 | SF |
| 1 | 0 | 1 | 0 | 1 | 1 | SS |
| 1 | 1 | 0 | 1 | 1 | 1 | FS |

# Corner: TT

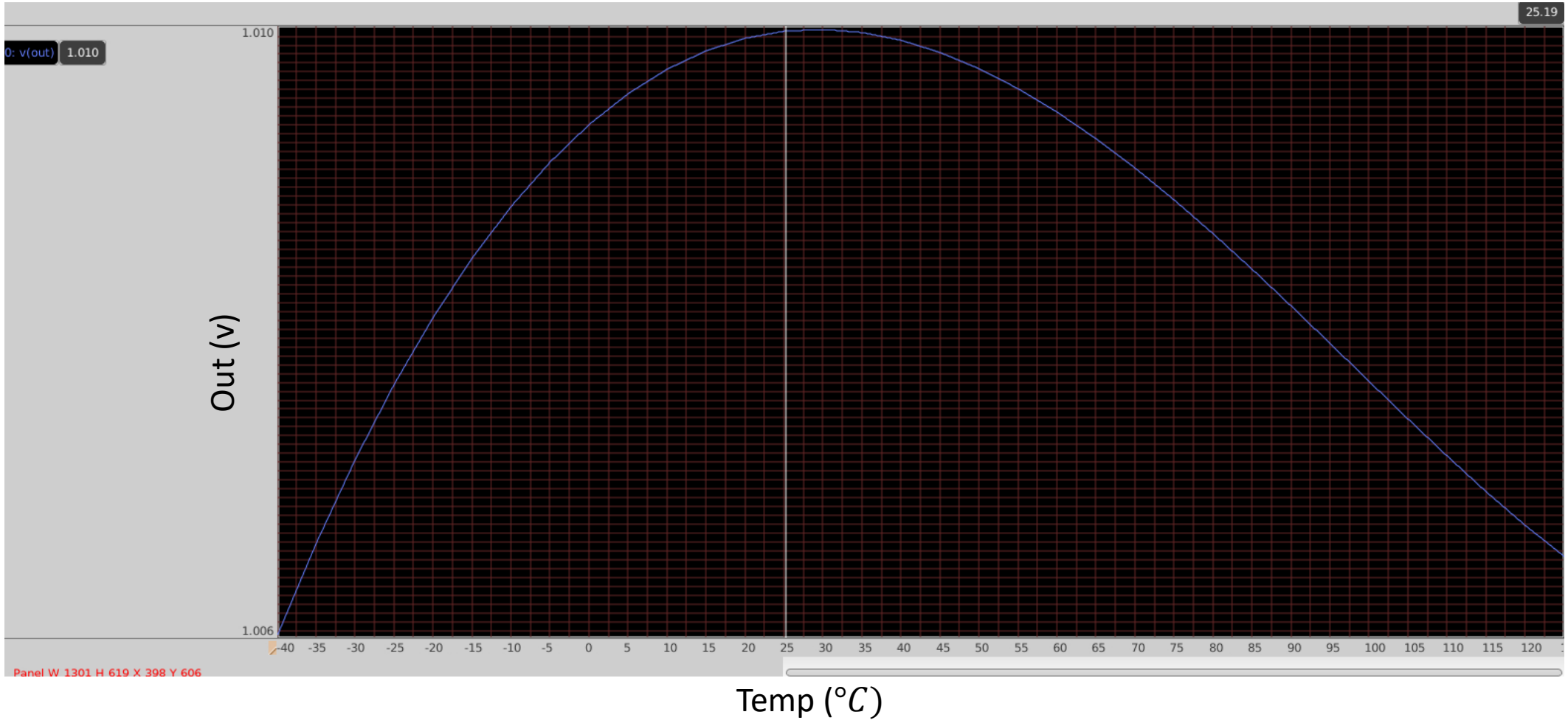



**Figure 6:** Output voltage versus temperature at the TT corner

$$\text{Tc} = \frac{Vout,\max - Vout,min}{Vout,nom \times \Delta T} \times 10^6 = \frac{1.010 - 1.006}{1.010 \times (125+40)} \times 10^6 = 24\ ppm/°\text{C}$$

# Corner: FF

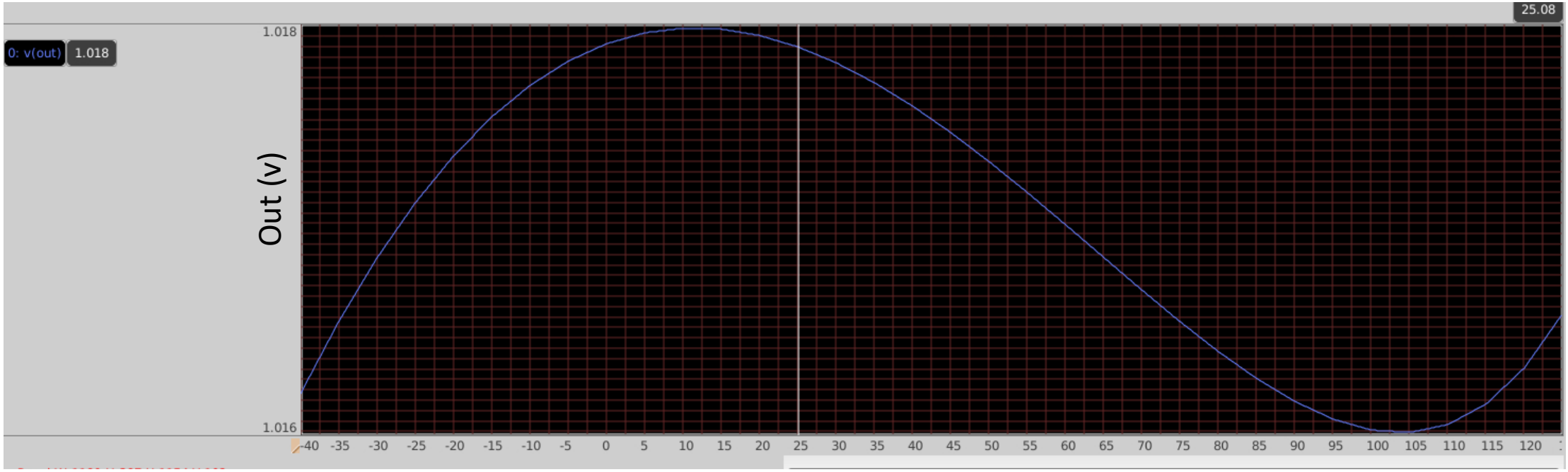



**Figure 7:** Output voltage versus temperature at the FF corner

$$\text{Tc} = \frac{Vout,\max - Vout,min}{Vout,nom \times \Delta T} \times 10^6 = 11.91\ ppm/°\text{C}$$

# Corner: FS

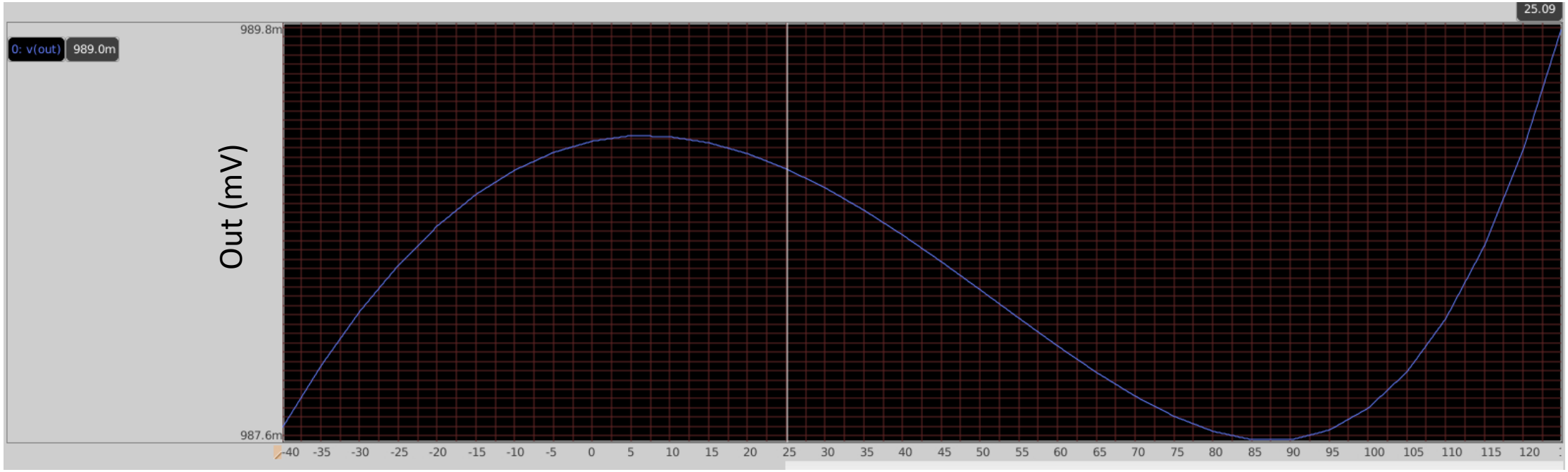



**Figure 8:** Output voltage versus temperature at the FS corner

$$\text{Tc} = \frac{Vout,\max - Vout,min}{Vout,nom \times \Delta T} \times 10^6 = 13.48\ ppm/°\text{C}$$

# Corner: SF

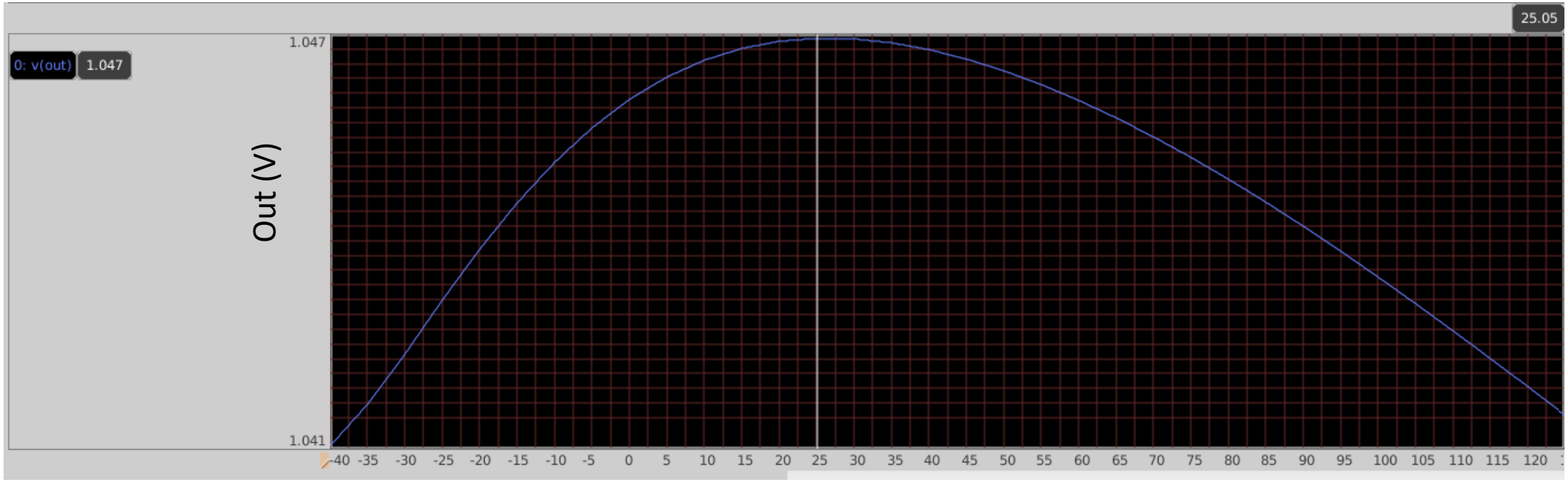



**Figure 9:** Output voltage versus temperature at the SF corner

$$\text{Tc} = \frac{Vout,\max - Vout,min}{Vout,nom \times \Delta T} \times 10^6 = 34.38\ ppm/°\text{C}$$

# Corner: SS

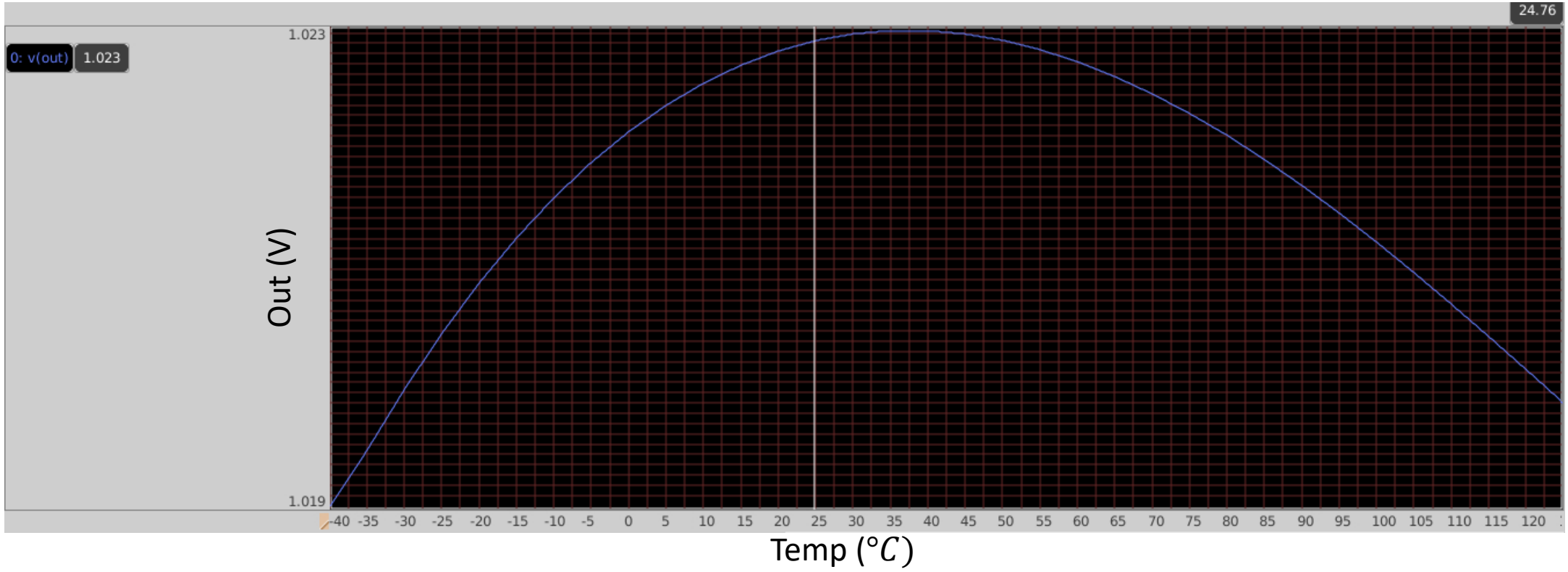



**Figure 10:** Output voltage versus temperature at the SS corner

$$\text{Tc} = \frac{Vout,\max - Vout,min}{Vout,nom \times \Delta T} \times 10^6 = 23.69\ ppm/°\text{C}$$

# TB_Stability

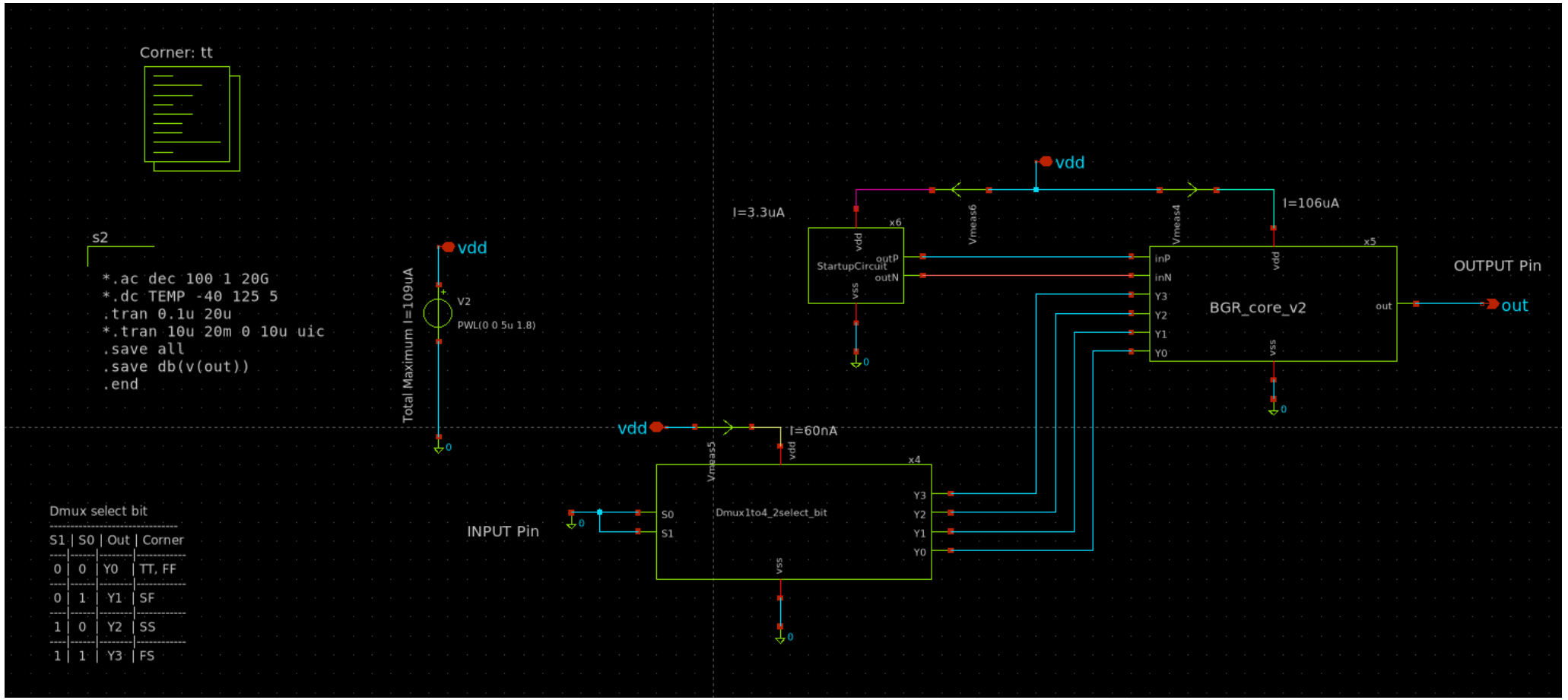



**Figure 11:** Stability testbench (TB_Stability)

# Stability_5us Ramp

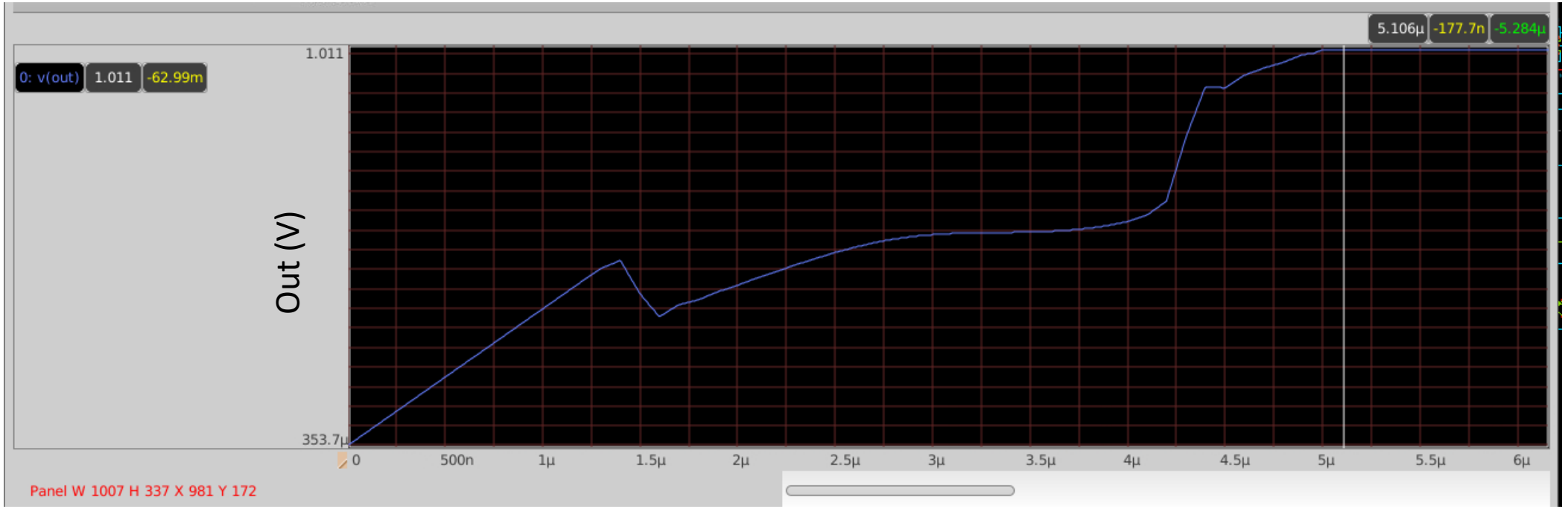



**Figure 12:** Output response to a 5 µs VDD ramp

For 5us Vdd ramp, the output is stable after 5.106us

# Stability_1us Ramp

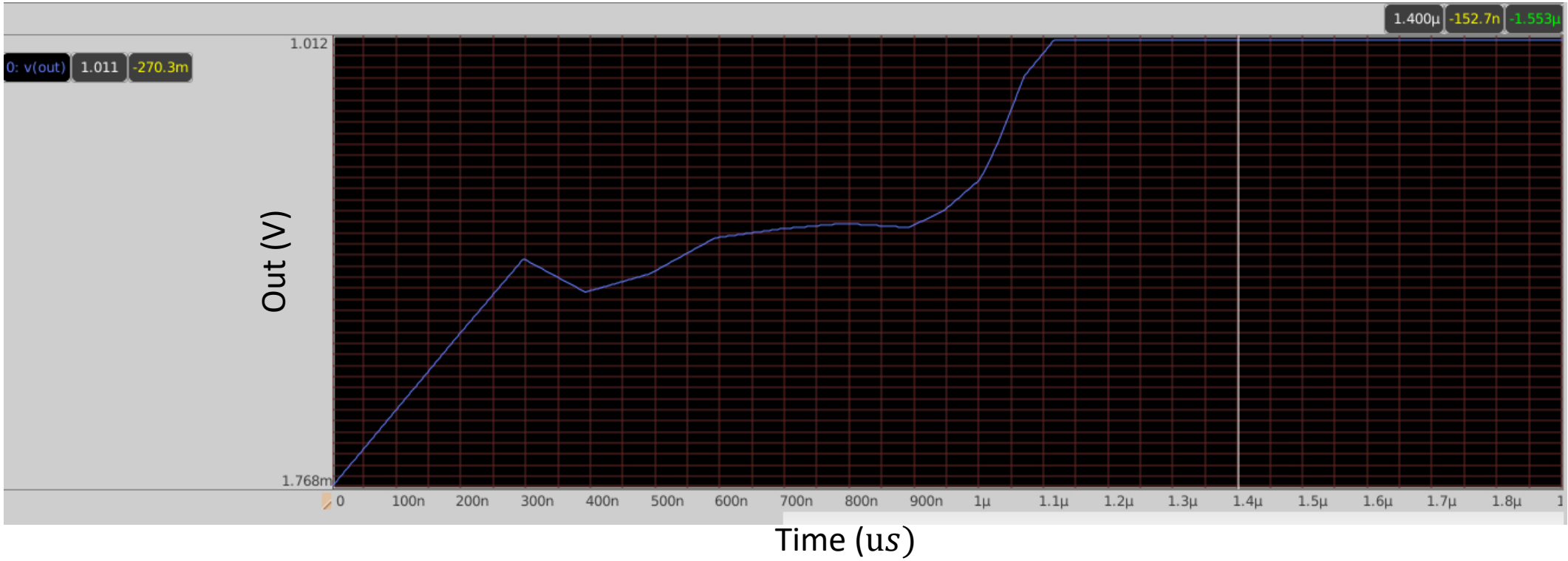



**Figure 13:** Output response to a 1 µs VDD ramp

For 1us Vdd ramp, the output is stable after 1.40us

# TB_PSRR

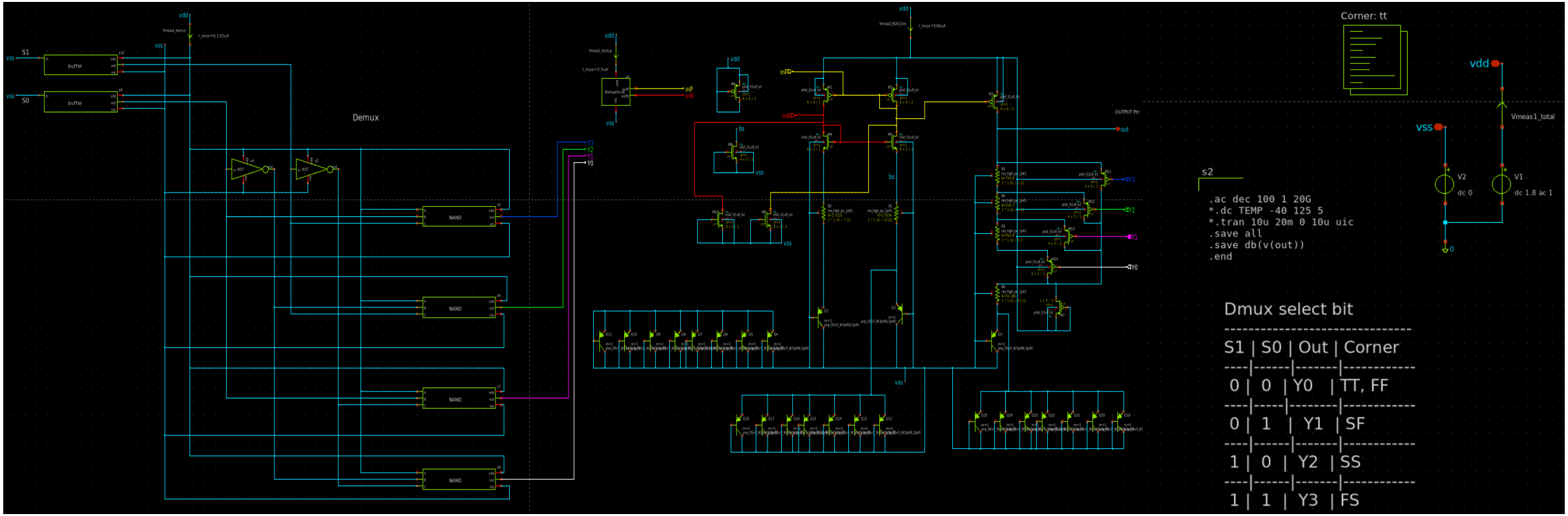



**Figure 14:** PSRR testbench (TB_PSRR)

# PSRR

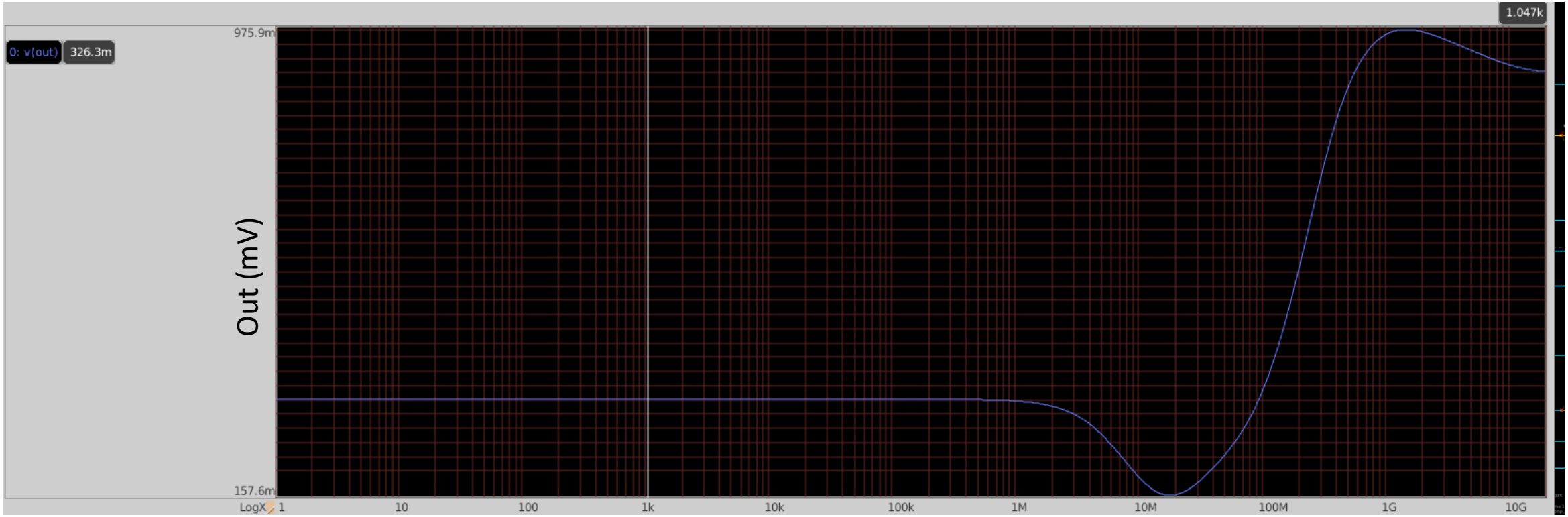



**Figure 15:** PSRR simulation: output response versus frequency

$$\text{PSRR}=20\log\left(\frac{Vdd}{Vout}\right) = 20\log\left(\frac{1}{0.3263}\right) = 9.73\text{dB}$$

# Key Results at Different Corners

**Table 3:** Key results at different corners: schematic vs. post-layout (PEX)

| Corner | Schematic | | | | Post-layout (PEX) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Tc (ppm/°C) | Trim R6 (Ω) | Vout @ 27°C (V) | Power (µW) | S1 (V) | S0 (V) | Tc (ppm/°C) | Vout @ 27°C (V) | PSRR @ 1 kHz (dB) | Power (µW) |
| **TT** | 24 | 11.526k | 1.010 | 192.96 | 1.8 | 0 | 26.6 | 1.0213 | 9.47 | 193.72 |
| **FF** | 11.91 | 11.526k | 1.018 | 198.36 | 1.8 | 0 | 21.4 | 1.0241 | 9.54 | 199.89 |
| **SS** | 23.69 | 10.866k | 1.023 | 187.56 | 1.8 | 0 | 32.5 | 1.0185 | 8.99 | 188.11 |
| **FS** | 13.48 | 10.18k | 0.9890 | 197.28 | 1.8 | 1.8 | 25.3 | 0.9887 | 7.66 | 199.20 |
| **SF** | 34.73 | 11.109k | 0.9985 | 188.82 | 0 | 1.8 | 25.3 | 1.0407 | 11.19 | 189.57 |

R6_trim base is 10.18 kΩ; Rds of each switch MOS is added in series with each selection. S1/S0: 1.8 V = logic 1, 0 V = logic 0.

# LVS Verification

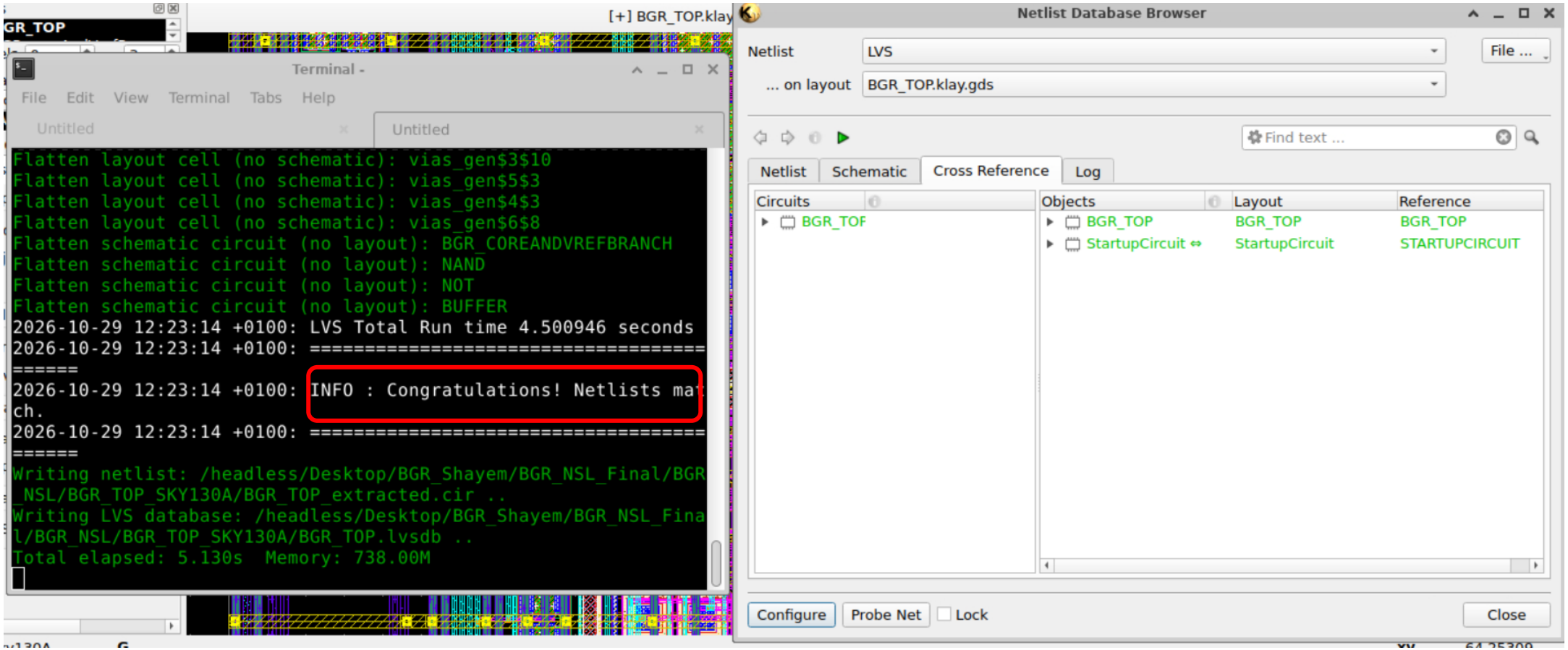



**Figure 16:** LVS verification result

# DRC Verification

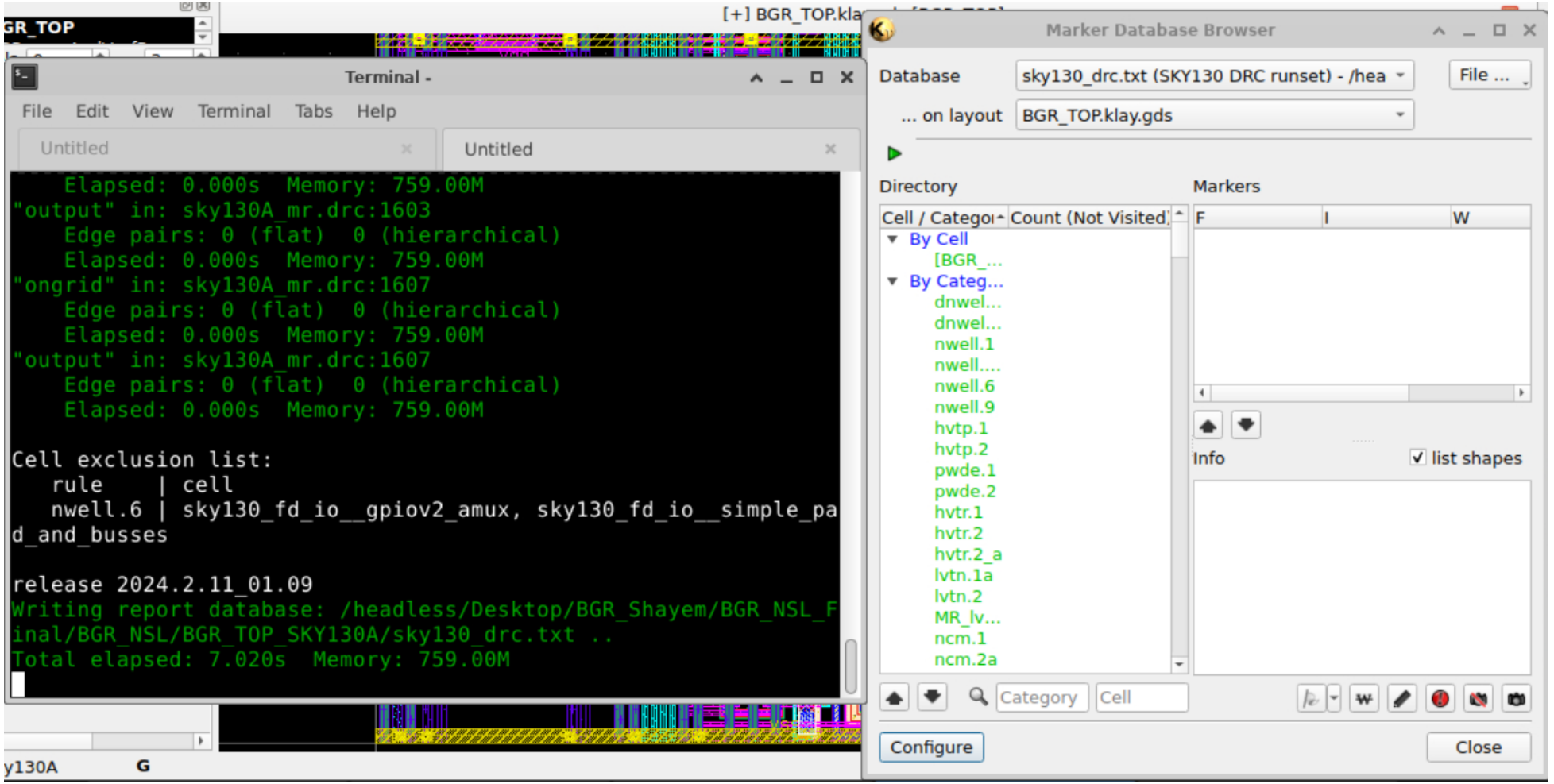



**Figure 17:** DRC verification result

# Top Level Layout

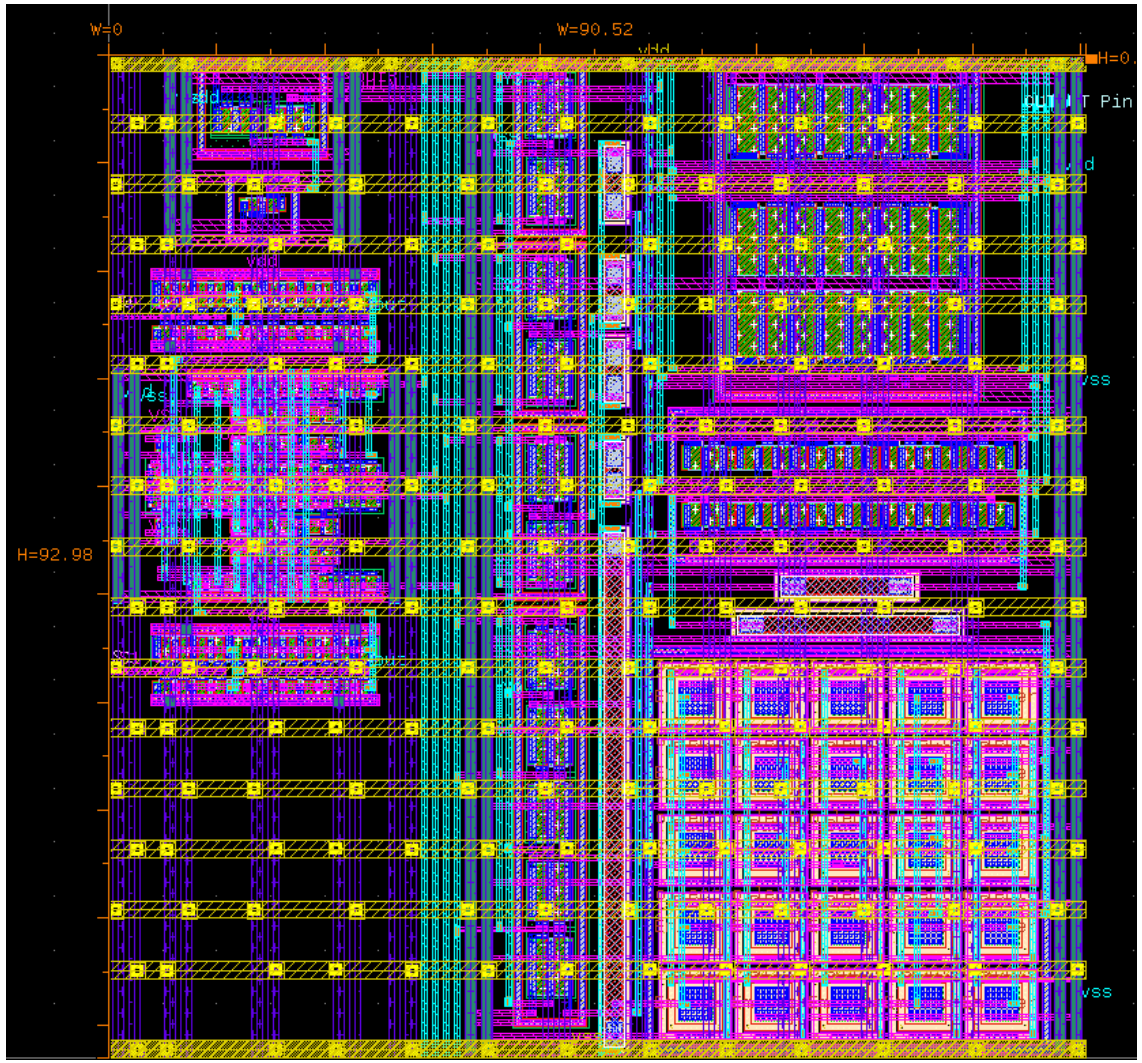



**Figure 18:** Top Level Layout with area 92.98 um X 90.52 um

# Access The Testbench to Reproduce

- Download the file from GitHub or go to the folder “BGR_NSL/BGR_TOP_SKY130A“ and find “BGR_TOP.sch” and BGR_TOP.klay.gds for LVS and DRC.

- For Schematic simulation, find “TB_BGR_TOP.sch” and change the select bit as shown in the TB or Demux select bits (Table 2) shown earlier in this document. Testbench for PSRR and Stability also be found as “TB_BGR_TOP_PSRR.sch” and “TB_BGR_TOP_Stability.sch”, respectively.

- GitHub Link: https://github.com/AShayem/BGR_NSL_FINAL.git

# SKY130 PDK Schematic and Layout Mismatch

- Resistance shows in the schematic and effective resistivity is different. Effective resistance in schematic is measured by $R=V_R/I_R$
- In layout, effective resistance is around 30 ohms higher than schematic's effective resistance.
- In layout, 0.12um is added by default, so 0.12um should be adjusted in schematic to resolve LVS.

*Effective resistance in layout is calculated after PEX.